Total Number of Pages 3 Roll No.....

B.Tech. Electrical Engineering *5th SEMESTER*

END SEMESTER EXAMINATION *(Nov. 2023)*

EE-301-Power Electronics

Time: 3:00 Hours ***Maximum Marks:40***

Note: Assume suitable missing data, if any.
Attempt all parts of a question at one place *(Marks may not be awarded otherwise)*
Q.1 and Q.2 are *COMPULSORY* attempt other three from the rest

1 Giving reasons by explaining (*very briefly*), why? Answer any **8(Eight)**. CO 1-5

(a) The thyristors are not transported without keeping anode and cathode terminal shorted.

(b) Only fast recovery diodes are advocated for freewheeling applications.

(c) AC-DC voltage doublers are not advocated for high power applications.

(d) Only 12n ±1 harmonics can exist in current for 12 pulse rectifier.

(e) 3 phase-controlled bridge rectifiers transact quasi square current for the mains.

(f) Filter requirements for inverter reduces for bridge inverters with Unipolar switching as compared to Bipolar switching.

(g) Single Phase Quasi square (120° – central aligned) wave inverters do not have triplens (*3rd harmonics and their multiples*).

(h) DC-DC Boost converters are usually not recommended for operation beyond a gain of 3.

(i) The requirement of capacitor at output of DC-DC boost converter is higher than that of buck converter.

(j) The range of the operation of the antiparallel thyristors connected in tandem with pure inductor fed from sinusoidal ac source is between 90°-180° to realize a variable inductor.

(k) Voltage stress of the switch used in the boost converter is very high as compared to that of buck converter. **1x8**

2 [a] For the thyristorised converter having a freewheeling diode in shunt with the load, determine the average value of the voltage across the load, the average load current, and the average values of the thyristor and the diode currents if the load is pure resistance. **2** CO 1,2,4

[b] An AC/AC voltage controller (*two thyristors in anti-parallel connection*) fed from 220 V, 50 Hz ac voltage source is loaded by a resistor R_L= 2.2 Ω. If the heater load is drawing power of 11kW, determine the angle of conduction of the thyristors. *[Given: value of* ***sin*** *for π/6=0.5; for π/4=0.707; for π/3=0.866 and for π/2=1]* **2**

[c] Draw the voltage waveforms of *1 Ph* full thyristor converter with load having high *L/R* ratio for $\alpha=30°$ and $\alpha=120°$ 2

[d] A single diode is used to rectify the current from an ac source of voltage 1000V at 2kHz to a resistive load of 1Ω. For the conditions prevailing and diode specifications, reverse recovery charge $Q_{rr}=10\mu C$ and softness factor S=0.6. Find reverse recovery time(t_{rr}) and peak reverse recovery current (I_{rr}) during commutation. 2

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3 [a] The full-wave rectifier having capacitor filter feeding a resistive load (500Ω) has a 230V source at 50 Hz. Determine the value of capacitance that would reduce the output voltage ripple to 1% of the dc value. 2

CO 1-2

[b] Determine the average value of the load current drawn by a *1Ph* thyristorised rectifier fed from 230V,50Hz AC source having source inductance (L_s) of 10mH, when thyristors are fired at $\alpha=\pi/3$. Assume $\omega L >> R$, to affirm constant current loading, while supplying to connected load(R=10 Ω, L large value). 3

[c] For a circuit having MOSFET with junction to case thermal resistance of 1.87°C/W, thermal resistance of case to heat sink of 0.5°C/W and thermal resistance of heat sink to ambient 7.2°C/W, determine maximum power that can be observe without exceeding junction temperature of 150°C, where ambient temperature is 40°C. In case the heat sink limits the junction temperature of 150°C for absorption of 15W, compute the thermal resistance for such heat sink. 3

4 [a] Derive the expression for the load voltages (v_{an}, v_{bn}, v_{cn}) of the three phase thyristorised inverter for 180° conduction. Draw a neat diagram to support the analysis. 4

CO 3,5

[b] Design a boost converter to provide an output of 18 V from a 12V source. The load is 20 W with switching frequency 60kHz. The output voltage ripple must be less than 0.5 percent. Specify the duty ratio, the inductor size and rms current rating, and the capacitor size and rms current rating. Design for continuous current. Assume ideal components. 4

5 [a] Design a buck converter for continuous conduction with the following specifications:

CO 3,5

V_{in} =24V(*in a range of [20V- 28 V]*), V_{out} =12V, and P_{out} =50W (*in a range [12W - 120 W]*, operated at the switching frequency of 100 kHz. Assume 85% efficiency. 4

[b] A single phase full bridge Inverter feeding a RL load(L=25mH,R=10Ω) with square and quasi-square voltage. If V_{DC}=100V and f=50 Hz, compare the total harmonic distortions of the voltage for square and quasi-square voltage (*where pulse is chopped off symmetrically in each half & on each side by 30°*). 4

6 [a] The two top switches of a *1Ph* full bridge inverter are switched with fundamental frequency of 50Hz and the other two lower switches with switching frequency, thereby realizing unipolar PWM. The DC bus is kept at 400V, terminal voltage of the inverter 223V AC and rms current output of 22A. Compute the rms currents experienced by top and bottom switches. 4

CO 2,5

[b] Derive the equation for current drawn by a 12 pulse AC-DC converter realized by accumulating the currents of two 3 phase rectifiers (30° phase displaced) with respect to each other. Use fourier series to derive the equations of currents. Assume quasi-square shape of the current. Show clearly that all triplens and other harmonics except 12n±1 vanishes (*where n=1,2,3....*). 4

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5th SEMESTER
B.TECH.

NOV/DEC-2023

END TERM EXAMINATION

EE-303 POWER TRANSMISSION AND DISTRIBUTION

TIME 3:00 Hours

Max. Marks: 40

NOTE: Attempt any five questions.
All questions carry equal marks.
Attempt all parts of a question at the same place.
Assume suitable missing data, if any.

Q.1 a) Explain different types of HVDC links. [CO6][2]

b) Calculate the volume of conductor material required for single phase two wire AC system. [CO1][2]

c) What is Corona loss? How it can be determined? [CO3][2]

d) What is the significance of GMD and GMR? [CO2][2]

Q.2 a) Explain the significance of propagation constant and characteristic impedance of a transmission line. [CO3][2]

b) A single circuit 50 Hz transmission line is 230 km long. The parameters of the line are given as $z=0.1603+j0.8277\ \Omega/km$ and $y= 5.105\angle 90°\ \mu\mho/km$. The load on the line is 125 MW at 215 kV and unity power factor. Find the ABCD parameters of transmission line using rigorous method considering long transmission line. Also calculate propagation constant and characteristic impedance of this line. [CO2][6]

Q.3 a) What are the factor which affect Corona? [CO3][4]

b) Each conductor of a three phase high voltage transmission line is suspended by a string of four suspension type disc insulators. If the potential difference across the second unit from the top is 13.2 kV and across the third unit from the top is 18 kV. Determine the line voltage of the transmission line and also the string efficiency. [CO5][4]

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Q.4 a) Explain the sag calculation for transmission line having towers of unequal height. The base of towers may be assumed to be at same level. **[CO5][4]**

b) An overhead transmission line is supported by towers of equal height and at the same base level, having a span of 250 meters. Calculate the sag under a wind pressure of 400N/m^2 and ice covering of 1 cm thickness. The conductor diameter is 1.1 cm and its weight is 650 kg/km. The breaking load and factor of safety are 7000 kg and 2 respectively. The ice density is 915 kg/m^3. One kg mass is equal to 9.81 N **[CO5][4]**

Q.5 a) Explain in detail the construction of a high voltage cable. **[CO4][4]**

b) A single core lead covered cable is to be designed for 66 kV to earth. Its conductor radius is 0.5 cm. The space between core and sheath is filled with three insulating materials. The relative permittivities of the three insulating material are 4, 4 and 2.5 respectively with the maximum stress 50, 40 and 30 kV/cm respectively. Determine the internal diameter of lead sheath.

[CO4][4]

Q.6 a) Considering a short transmission line, explain the effect of load power factor (lagging, UPF and leading) on the voltage regulation of transmission line. **[CO2][4]**

b) Calculate the inductance of a three phase double circuit transmission line as shown in Fig. 1. The conductor self GMD is 0.467 cm. Assume symmetrical spacing of conductors. **[CO2][4]**

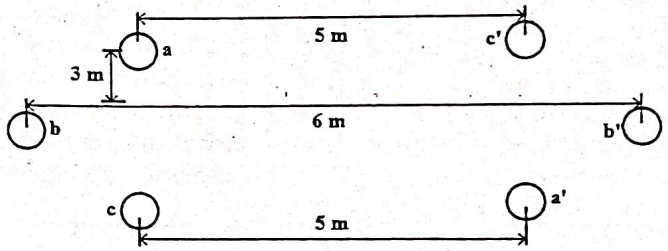


Fig. 1

Q.7 Write short notes on any two.

a) Surge arrestor **[CO3][4]**

b) Capacitance of three core cable **[CO4][4]**

c) Types of insulators **[CO5][4]**

Total no. pages: 03 *Roll No.....*

5th and 7th SEMESTER

B.Tech.

END TERM EXAMINATION ***Nov/Dec-2023***

EE319 Digital System Design

Time: 3:00 Hours *Max. Marks: 40*

Note :
If necessary, make appropriate assumptions and approximations and state them
You must solve all the questions and their parts in the serial order

Q.1 Answer any FIVE. [3X5=15]

(a) Determine the probability that an energy level 3 kT above the Fermi energy is occupied by an electron at T = 300K. [CO#1]

(b) Explain vacancy and interstitial defects in a crystal. [CO#1]

(c) Explain how dynamic CMOS design technique achieves power savings compared to static CMOS design technique during the input transitions. [CO#2]

(d) The time period of the clock signal (clk) generated by the following code segment will be _____ time units. (Please provide numeric answer with justification).

```
intial
#15 clk = 1'b0;
always
#12 clk = ~ clk;
```

[CO#4]

(e) "PMOS transistor cannot pass strong logic 0". Justify the statement. [CO#2]

(f) Why TSPC latches are preferred as compare to Static D latches? [CO#3]

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Q2. Answer any Two. [5X2=10]

(a) Use gate level description of Verilog HDL to design a 4X1 MUX. Write truth table, logical expression and logical diagram. [CO#4]

(b) Describe the following statements in Verilog HDL: initial and always. Also design a clock with period = 40 and a duty cycle of 25% by using the initial and always statements(in Verilog). [CO#4]

(c) What is the function of the circuit in Figure 1? (with proper justification) [CO#3]

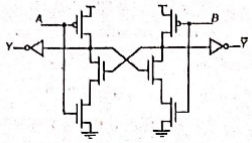


Figure 1

Q3. Attempt any two [3X2 = 6]

a) What logic equations do the following schematics implement in Figure 2? [CO2]

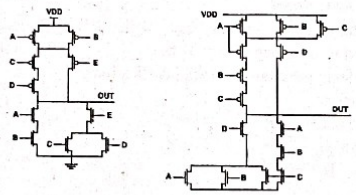


Figure 2

b) What is the output function of the following circuit show in Fig. 3? [CO#2]

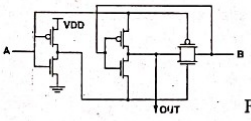


Figure 3

c) Differentiate TSPC latch with C²MOS latch. [CO#3]

Q3. Figure 4 contains a pass-gate logic network. [3] [CO#2,3]

(a) Determine the truth table for the circuit. What logic function does it implement?

(b) If the PMOS were removed, would the circuit still function correctly? Does the PMOS transistor serve any useful purpose?

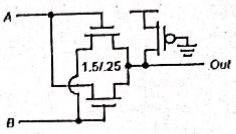


Figure 4

Q6. Consider Figure 5. Ignore all capacitances at the "out" node: [6][CO#2]

(a) When input signal (IN) switches from high to low, which transistor would first turn from ON to OFF state?

(b) What are the regions of operation of transistors Mp1 – Mp3 and Mn1 – Mn3 for Vin = 0.7V (Vth = 0.3V) during the high to low input transition?

(c) What is the value of the current through Pull up network?

(d) During the low to high transition, which of Mn1 and Mn2 first turns ON?

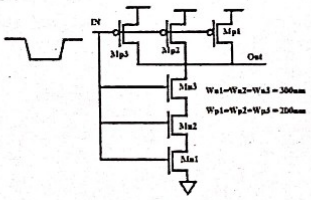


Figure 5

Total No. of Pages: 03 Roll No. .....

FIFTH SEMESTER B.Tech.[EE]

END-SEMESTER EXAMINATION (Nov. 2023)

EE-323 FUNDAMENTALS OF MACHINE LEARNING

Time: 3 Hour *Max. Marks: 40*

Note: Attempt "ANY FOUR" Questions. Assume any data if missing and clearly mention the assumption

Q1. a) Explain why dimensionality reduction is useful in machine learning? What are the commonly used dimensionality reduction techniques in machine learning? Explain any one in brief. **[5] [CO-1]**

b) Determine the Singular value decomposition (SVD) of the following matrix "A" **[5] [CO-2]**

$$A=\begin{bmatrix} 1 & 0 & 1 \\ -2 & 1 & 0 \end{bmatrix}$$

Also, comment on how the eigenvalues and singular values are related?

Q2. a) Explain k-means algorithm with a pseudo code. Also discuss some of its drawbacks. **[5] [CO-1]**

b) Use k-means clustering algorithm to divide the following data into two clusters:- **[5] [CO-2]**

x_1	1	2	2	3	4	5
x_2	1	1	3	2	3	5

Q3. a) Generate a Decision Trees that fits the below-mentioned data **[10] [CO-3]**

for classifying whether a match will be played or not :

Instances	Outlook (x_1)	Temperature (x_2)	Humidity (x_3)	Play Tennis (y)
1.	Sunny	Hot	High	No
2.	Sunny	Cool	Normal	Yes
3.	Overcast	Hot	High	Yes
4.	Rain	Mild	High	Yes
5.	Rain	Cool	Normal	Yes
6.	Rain	Mild	Normal	Yes
7.	Overcast	Cool	Normal	Yes
8.	Sunny	Mild	High	No

Q4. a) Consider the data given in the table below and classify a sample x={M, 1.95} using Naïve-Bayes classifier. **[10]** **[CO-4]**

Sample No.	Gender (x_1)	Height (x_2)	Class
1.	F	1.6 m	Short
2.	M	2 m	Tall
3.	F	1.9 m	Medium
4.	F	1.88 m	Medium
5.	F	1.7 m	Short
6.	M	1.85 m	Medium
7.	F	1.6 m	Short
8.	M	1.7 m	Short
9.	M	2.2 m	Tall
10.	M	2.1 m	Tall
11.	F	1.8 m	Medium
12.	M	1.95 m	Medium

Q5. a) Discuss why the **logistic function** is an appropriate hypothesis function for logistic regression, particularly in comparison to a linear hypothesis function. Also justify why the **Binary Cross-entropy Loss** is preferred over **MSE (Mean Squared Error)** for logistic regression despite MSE being used in linear regression. **[5]** **[CO-5]**

b) Find the first derivative of the log-likelihood for logistic regression with one predictor variable. Explicitly write out the formula for doing one step of gradient descent method. **[5]** **[CO-4]**

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Total No. of Pages 02

Roll No. 213

V / VII SEMESTER

B.Tech. [EE]

END-SEM EXAMINATION

DEC-2023

EE325 INTERNET OF THINGS

Time: 3 Hour

Max. Marks: 50

Note: Attempt any five questions. All questions carry equal marks

Q.1[a] Compare cyber physical systems versus IoT. [5] CO2 L2

[b] Mention clearly the procedure to interface the heartbeat sensor with any microcontroller of your choice. [5] CO1, CO4 L2

Q.2[a] Compare classic Bluetooth with BLE (V4.2). [3] CO4 L2

[b] Enlist and discuss in brief the four types of messages supported by CoAP. [5] CO2 L2

[c] IoT belongs to IT domain. Share your views. [2] CO3 L3

Q.3[a] Discuss the various types of MQTT schemes. [4] CO2 CO5 L2

[b] What do you understand by Li-Fi protocol. Explain its functioning, advantages and disadvantages [6] CO2 CO5 L3

Q.4[a] One of the PC on DTU network shows the IP address as 10.13.2.224. Which version of IP address is this? What should be its subnet mask? What is the class of this IP address and justify the use of this class of IP address for DTU network. [6] CO3 L2 L3

[b] Discuss type of service field as applied to IPV4 format. [4] CO3 L3

Q.5[a] Discuss and compare the three deployment models related to cloud computing. [5] CO2 L2 L3

[b] Enlist and discuss in brief the challenges related to IoT with cloud [5] CO5 L3

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Q.6 [a] Discuss the various criteria one should consider while selecting a cloud service provider. [5] CO5 L2 L3

[b] Discuss the model building process as applied to machine learning models. [5] CO5 L2 L3

Q.7[a] Discuss how IoT can be used to monitor the quality of perishables and vegetables during transport. [5] CO6 L2 L3

[b] How IoT can be used for elderly fall detection? Discuss the end-to-end solution. [5] CO6 L2 L3

Q.8 Write short notes on any two. [2x 5] All CO's

i. Application areas of IoT.
ii. Components of BLE
iii. Uniform resource identifier
iv. Difference between edge and fog computing.
v. IPV6

salvage value (money received after selling the machine at the end of its operational life)

	Alternative A	Alternative B
Initial Investment (in Rs.)	1,50,000	1,75,000
Annual Equal Return (in Rs.)	60,000	70,000
Salvage value (in Rs.)	15000	35000

Determine the best alternative based on the annual equivalent method when rate of interest is 25%.

7 a What do you understand by Recession? Discuss its effect on producers, employees of a manufacturing firm and students. [2+3=5][CO#2]

b Excel manufacturing firm to expand its production operation. It has identified three different technologies for meeting the goal. The initial investment and annual expenditure with respect to each of the technologies is given below. [5][CO#3]

	Initial Investment (in Rs)	Annual Maintenance cost	Life (in years
Technology 1	12,00,000	4,00,000	10
Technology 2	20,00,000	6,00,000	10
Technology 3	18,00,000	5,00,000	10

Suggest the best technologies which is to be adopted based on the present worth method of comparison assuming 20% interest rate compounded annually.

Total No. of Pages-4 *Roll No. .....*

Vth Semester

B.Tech.

End Term Examination **Nov./Dec.- 2023**

HU 301 Engineering Economics

Time- 3.00 hrs **Marks- 50**

Note: 1. Answer any five questions.
2. All questions carry equal marks.
3. Assume missing value, if any

1 a Discuss salient feature of the Indian economy. [5][CO#1]

b The accounts of ABC manufacturing LTD. for the year ended 31st March 2023 shows following: [5][CO#2]

Particular	Rs.
Stock of Material on 01.4.2022	87,200
Materials purchased	369000
Drawing office salaries	40000
General office salaries	15000
Bad debts written off	12000
Salesperson's salary & commission	17000
Depreciation written off of furniture	10000
Rent, tax and insurance of factory	41900
Productive Wages	176400
General expenses	40000
Gas and water (factory)	2680
Travelling expenses	2680
Managers Salary	75000
Depreciation of plants and tools	9,100
Cash discount allowed	4060
Repair pf plats and tools	6230
Carries outwards	6030
Direct expenses	10010
Rent, taxes, insurance (office)	2800
Gas, water (office)	560
Stock of material 31.3.2023	87920

Prepare cost seat for the year ending 31.3.2023

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2 a Think one product of the Indian market whose market is oligopoistic and one product whose market is monopolistic competition. Discuss reasons for your decision. [5][CO#3]

b A mall owner is taking stock of his expenditure on advertisement. It has increased its expenditure from 50 thousand to Rs. 75 thousand each on advertisement of commodity X1 and X2. Statistics on its pre and post advertisement sell is given below: [5][CO#1]

Commodity	Pre-advertisement sell	Post advertisement sell
X1	3 lakh	3.12 lakh
X2	5 lakh	7.70 lakh

Find out their advertisement elasticity and nature of X1 and X2.

3 a You are CEO of a company and you have to decide a foreign location where you would like to expand your business. Discuss factors you would like to consider for taking decision. [5][CO#3]

b If price of A changes from 20 to 25 per unit, following are the detail of demand of A, B and C. Calculate the Cross Elasticity of Demands and find the relation between the commodity A, B and C. [5][CO#2]

Commodity	Demand	
	Original	Changed
A	100	80
B	120	150
C	200	160

4 a What do you mean by opportunity cost? How it is different from actual cost? Discuss with examples [2+3=5] [CO#2]

b A small business with an initial investment of Rs. 12,000 yields Rs. 10,000 during the first year of operation and the yield increases by Rs. 1,000 from its second year of operation up to its 10th year of operation. At the end of life of the business, the salvage value is zero. Find the present worth of the business by assuming an interest rate of 18%, compounded annually. [5][CO#2]

5 a You are Governor of the Central Bank, country ABC. The inflation rate has become 9. Is it a matter of concern? What fiscal and monetary measures may be taken and why? Discuss with examples. [5][CO#3]

b A man owns a corner plot. He must decide which of the two alternatives to select in trying to obtain a desirable return on his investment. After much study and calculation, he decides that there are two best alternatives. Details are given below: [5][CO#2]

	Build Gas station	Build Ice-cream stand
Initial Investment (in Rs)	20,00,000	36,00,000
Annual Property Tax (in Rs.)	80,000	1,50,000
Annual Income (in Rs)	8,00,000	9,80,000
Life of Building (in Years)	20	20
Salvage Value (value which owner may receive after completing life) (in Rs.)	0	0

Evaluate the alternative based on the future worth method.

6 a What do you mean by Online work? Is it good or bad type of employment? Whether engineering students should be encouraged to do it? Discuss with examples. [5][CO#3]

b A company invests in one of the two mutually exclusive alternatives. The life of both alternatives is estimated to be 5 years with the following initial investments, annual return and [5][CO#2]

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